

Relation Between Higher Physical Activity and Public Transit Use

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Physical inactivity is prevalent and a worldwide public health concern.¹ Increasing active transport is an appealing strategy to increase overall physical activity, although more clarity is needed about the amount of physical activity directly attributable to transportation choices. Users of public transit (e.g., bus, train) engage in more overall physical activity than do nonusers and more often meet daily physical activity recommendations (≥ 30 min/d on most days), likely from the active transport involved with accessing transit (e.g., walking to a bus stop).^{2–5} Reported total walking time is also higher among transit users than among nonusers.^{4,6} National travel diary data indicate that the average American transit user (approximately 2%–3% of adults) walks 19 minutes per day to and from transit, and approximately one third of these transit users attain recommended levels of physical activity based solely on the amount of walking related to their transit use.^{2,3} Better public transit access and more use appear related to more overall physical activity and specifically walking.⁷

However, many previous studies about the relation between transit use and physical activity fail to address the critical issues of possible confounding and potential substitution. Regarding confounding, examining only overall physical activity or total walking among transit users versus nonusers without disaggregating physical activity into constituent types and purposes of walking makes it difficult to determine how much physical activity is directly related to using public transit (i.e., walking or biking to and from public transit stops), separate from other types of utilitarian (e.g., walking to the store) or recreational physical activity. This is problematic because other factors could readily account for the relation between transit use and higher overall physical activity and walking. For example, built environment factors such as residential density and land use mix are related to transit use because transit access is higher in more

dense and higher mix areas; however, these built environment variables are also related to walking to and from other nonresidential locations (e.g., stores, restaurants).⁸ Thus, without more precision, it is not possible to rule out a spurious relation (e.g., through built environment or other shared variables) between transit access and use and physical activity.

The issue of substitution is also critical to measuring the health effect of transit use on physical activity. If transit users decrease the time spent in other activities in lieu of the time spent in transit-related walking, attempts to increase transit use would not yield increases in physical activity but merely shift from one form to another form of physical activity. Studies that provide more precise estimates of walking to and from transit use have not examined whether such substitution occurs. A recent US time use study suggested that some substitution may be happening as individuals with longer commutes, which are often characteristic of

public transit use, engage in less recreational physical activity than do those with shorter commutes.⁹

Large-scale randomized trials manipulating transit use are not feasible, but one approach to a better understanding of the relation between transit use and physical activity is to obtain an objective overall and type-specific assessment of physical activity among transit users and nonusers and, for the former, to examine walking behavior directly linked to transit use. For the current study, we hypothesized that transit users would be more physically active than nonusers and that increased activity would be directly linked to transit use (i.e., no or little confounding). Thus, we expected that physical activity would be higher among more frequent transit users than among less frequent users and that more physical activity would occur only on days when transit was used as a result of the transit-related physical activity and not changes in other types

Objectives. We isolated physical activity attributable to transit use to examine issues of substitution between types of physical activity and potential confounding of transit-related walking with other walking.

Methods. Physical activity and transit use data were collected in 2008 to 2009 from 693 Travel Assessment and Community study participants from King County, Washington, equipped with an accelerometer, a portable Global Positioning System, and a 7-day travel log. Physical activity was classified into transit- and non-transit-related walking and nonwalking time. Analyses compared physical activity by type between transit users and nonusers, between less and more frequent transit users, and between transit and nontransit days for transit users.

Results. Transit users had more daily overall physical activity and more total walking than did nontransit users but did not differ on either non-transit-related walking or nonwalking physical activity. Most frequent transit users had more walking time than least frequent transit users. Higher physical activity levels for transit users were observed only on transit days, with 14.6 minutes (12.4 minutes when adjusted for demographics) of daily physical activity directly linked with transit use.

Conclusions. Because transit use was directly related to higher physical activity, future research should examine whether substantive increases in transit access and use lead to more physical activity and related health improvements. (*Am J Public Health.* 2014;104:854–859. doi:10.2105/AJPH.2013.301696)

of physical activity (i.e., no or little substitution occurs).

METHODS

Participants in the current analysis were the baseline sample, with data collected from 2008 to early July 2009, for the Travel Assessment and Community study examining the effect of a new light-rail system that would be opened in July 2009. Participants were selected to reside proximal (< 1 airline mile; case) or distal (> 1 airline mile; control) from future light-rail stops but to be living within the same county (King County, WA) and living in areas with similar built environments (captured as residential density; housing type; home values; bus transit access; and proximity to a neighborhood retail center defined by at least 1 grocery store, 1 restaurant, and 1 other retail store in close proximity to one another) and similar census-based demographic characteristics (household income and race/ethnicity).¹⁰ Eligible households in identified areas were contacted via address and telephone information from marketing companies. Participants needed to be (1) at least 20 years old (to better ensure independent decision-making about mobility and transportation, relative to younger adults or children), (2) able to complete the travel log and survey in English, and (3) able to walk unassisted for at least 10 minutes.

Overall, 706 participants provided accelerometer, Global Positioning System (GPS), and travel log information. Among these, 5 participants had physical activity bouts each day that could not be classified as walking or nonwalking, and 8 additional participants did not complete any of the attitudinal and demographic survey. Thus, 693 participants were included in the current analyses (649 with complete demographic data), averaging 51.0 years old (SD = 13.2) and having a body mass index (BMI; defined as weight in kilograms divided by the square of height in meters) of 26.6 (SD = 6.3). Participants were 63.5% female and 79.3% White and non-Hispanic, and 70.3% reported being college graduates. Reported annual household income was less than \$50 000 (38.7%), \$50 000 to \$100 000 (40.5%), and greater than \$100 000 (20.7%). Participants had similar demographics to the census block group urban area demographics of

King County (based on 2000 US Census data), with the county being 77.9% White and non-Hispanic (among adults) and having a median household income of \$59 000. The current sample contained a higher proportion of women relative to the county (50.4% in county) and college graduates (40.2% in county) and was older (median age of 37.0 in county, although this estimate includes those younger than 20 years). Participants' job status was full time (53.1%), part time (23.8%), retired (16.9%), and unemployed (6.3%). Among participants, 8% reported being a student.

Data Collection

Eligible and interested participants were mailed an accelerometer (Actigraph GT1M; ActiGraph LLC, Pensacola, FL), portable GPS device (GlobalSat DG-100; GlobalSat WorldCom Corp, New Taipei City, Taiwan), and 7-day paper travel log. Participants also were provided a written or an online (based on their preference) attitudinal and demographic survey to complete. Soon after receiving these materials, participants were contacted by study staff to review procedures (e.g., how to wear the devices; how to charge the GPS device nightly) and asked to wear the accelerometer and GPS for 7 days during waking hours and to complete the travel log for these days. Accelerometer data were aggregated to 30-second epochs, and GPS devices were set to collect data at 30-second intervals. Participants mailed back the devices and travel log (and survey if written form) in a prepaid envelope.

Data Processing

The process by which accelerometer data were integrated with GPS and travel log information to identify walking and nonwalking physical activity bouts is described elsewhere in detail.¹¹ Briefly, bouts of 5 minutes or longer of accelerometer counts greater than 500 per each 30-second epoch were considered physical activity; these bouts were then considered to be walking depending on GPS speeds and temporal overlap or proximity to walking and other trips recorded in the travel log. Travel logs included the places and durations reported to be visited and the travel mode between places. To be considered in the current analysis, an assessment day had to have at least 1 place record in the travel diary and an accelerometer

wearing time of 8 hours or longer. Accelerometer periods of 20 minutes or longer with continuous zeros were considered as nonwearing times. For the current study, GPS was used only to examine speed of travel, in order to identify potential walking trips (2–6 km/h), and not used to identify location or travel mode. The integration of GPS and travel log for the identification of walking behavior is superior to use of travel log alone.¹¹ An assessment day may or may not have had GPS data. The final sample for the current analysis consisted of 693 participants and 4432 person-days (mean = 6.4 days/person; SD = 1.8; 84.1% of the sample [*n* = 583] had 5–8 assessment days).

Transit users were defined as having 1 or more transit days, with transit days defined as any day in which a transit trip was recorded in the travel log. Transit-related walking was a bout of physical activity classified as walking that overlapped with, or was within 10 minutes in time (not distance) of a transit trip or place, regardless of whether a walking trip was recorded in the travel log (schematic available as a supplement to the online version of this article at <http://www.ajph.org>). A transit place was defined as the place from or to which a transit trip was accessed ("bus stop," "transit station," "metro tunnel," and "park & ride"). There were 274 participants having 1 or more transit days (39.5%) and thus classified as transit users. Transit users had an average of 2.97 (SD = 2.05) transit days, which on average represented 46.3% (SD = 27.9) of their assessment days. Transit users and nonusers did not differ significantly in their number of assessment days.

Data Analysis

We treated the participant as the unit of analysis to compare demographics, average daily overall physical activity, average daily nonwalking physical activity, and average daily transit- and non-transit-related walking between transit users and nontransit users with the χ^2 test or analysis of variance. We used analysis of covariance along with the Bonferroni post hoc test to examine differences in these physical activity metrics across nontransit users and groups defined by tertiles of transit users' proportion of assessment days that were transit days. Differences in demographics across transit user groups (gender, household income,

TABLE 1—Overall Daily Physical Activity and Walking and Nonwalking Between Transit Users and Nontransit Users in the Baseline Travel Assessment and Community Study Sample: King County, WA, 2008–2009

Variable	Transit Users (n = 274), Mean (95% CI)	Nontransit Users (n = 419), Mean (95% CI)	P
Overall physical activity, min/d	46.0 (42.2, 49.8)	37.6 (34.5, 40.7)	.001
Walking, min/d			
Total	32.1 (29.3, 34.9)	21.6 (19.4, 23.9)	< .001
Transit-related	7.6 (6.9, 8.3)	0	NA
Non-transit-related	24.5 (21.8, 27.2)	21.6 (19.4, 23.9)	.1
Nonwalking physical activity, min/d	13.9 (11.8, 16.0)	15.9 (14.2, 17.7)	.15

Note. CI = confidence interval; NA = not available. Means (95% CIs) adjusted for number of assessment days.

education level, and being Hispanic vs non-White/non-Hispanic White) were covariates in this analysis. For each participant, physical activity metrics were averaged across transit and nontransit days. We then examined the association between physical activity metrics and transit use with participant day-level data (transit days vs nontransit days) as the unit of analysis in a linear mixed effects model, and we controlled for age, gender, race, employment status, household income, education level, and weight status. Statistical significance was set at $P < .05$, and all analyses were conducted in SPSS version 17.0 (SPSS, Chicago, IL) and Stata version 12.1 (StataCorp LP, College Station, TX).

RESULTS

No demographic or socioeconomic differences were found between transit and nontransit

users, except that transit users had a significantly lower BMI than did nontransit users (26.0 vs 27.0; $P < .05$).

Transit users had significantly more daily overall physical activity than did nontransit users when averaged across all their assessment days. Transit users' total walking time per day also was significantly higher than that of nontransit users. Yet average daily non-transit-related walking and nonwalking physical activity did not differ significantly between transit users and nonusers (Table 1).

In comparing nontransit users and groups of transit users by the proportion of assessment days taking transit, significant differences were seen in daily overall physical activity, total walking, and transit-related walking. Transit users with the lowest proportion of transit days ($\leq 30\%$ of assessment days being days they took transit) had significantly lower average overall physical activity, total walking, and

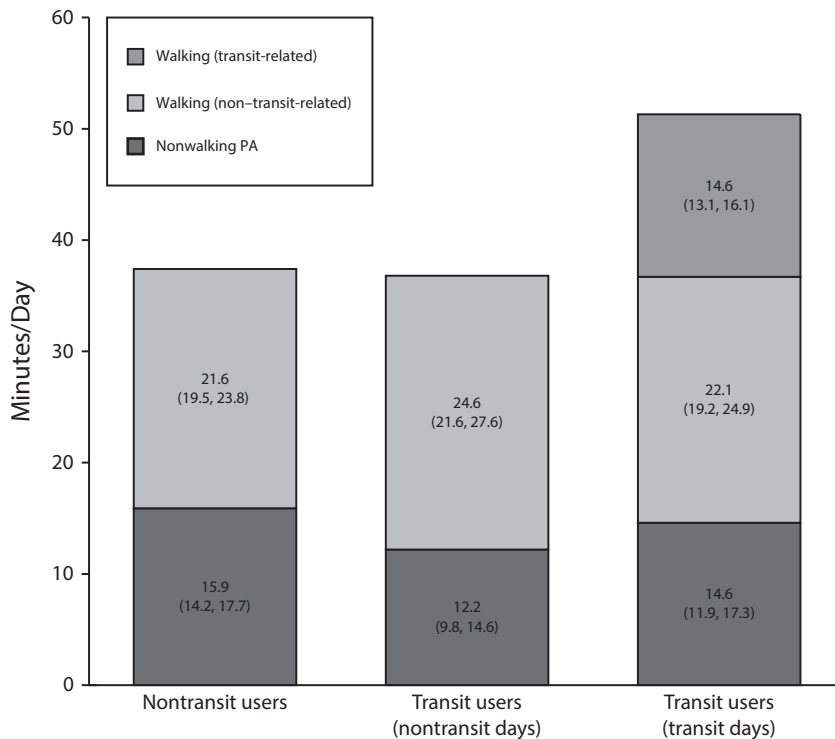
transit-related walking compared with those taking transit most often ($\geq 60\%$ of assessment days they took transit). The middle transit day proportion tertile (31%–59% of days being transit days) was not significantly different in overall physical activity or total walking from either the lowest or the highest tertiles but had lower average transit-related walking per assessment day than did those in the highest transit day proportion tertile (Table 2). There were no significant differences by transit day proportion tertile in non-transit-related walking or nonwalking physical activity. Transit-related walking was significantly lower among nontransit users compared with each of the transit day proportion tertiles, but nontransit users' overall physical activity was significantly lower than only those in the highest transit use tertile. Total walking also was significantly higher among the middle and highest transit day proportion tertiles than among nontransit users. All other comparisons with nontransit users were not significantly different.

When we examined activity at the day level, transit users engaged in significantly more daily total physical activity and total walking on transit days than on nontransit days (both $P < .001$). By contrast, non-transit-related walking ($P = .097$) and nonwalking physical activity ($P = .13$) did not differ between transit and nontransit days. Transit-related walking averaged 14.6 minutes on transit days (Figure 1). Taking transit on a day was related to approximately 12.4 more minutes of walking compared with days not taking transit, after demographics were included in the model (Table 3). Being male was also related to more total

TABLE 2—Overall Daily Physical Activity and Walking and Nonwalking by Nontransit Users and Tertiles of Transit Users' Proportion of Assessment Days That Were Transit Days in the Baseline Travel Assessment and Community Study Sample: King County, WA, 2008–2009

Variable	Nontransit Users (n = 394), Mean (95% CI)	Transit Use $\leq 30\%$ of Days (n = 99), Mean (95% CI)	Transit Use 31%–59% of Days (n = 73), Mean (95% CI)	Transit Use $\geq 60\%$ of Days (n = 83), Mean (95% CI)	Overall P
Overall physical activity, min/d	37.7 (34.6, 40.8) ^a	39.3 (33.1, 45.5) ^{a,b}	46.3 (39.1, 53.5) ^{a,b}	51.7 (44.8, 58.5) ^b	.001
Walking, min/d					
Total	21.8 (19.5, 24.0) ^a	25.8 (21.3, 30.4) ^{a,b}	34.4 (29.1, 39.7) ^{b,c}	36.5 (31.5, 41.5) ^c	< .001
Transit-related	0 ^a	2.3 (1.3, 3.3) ^b	6.5 (5.4, 7.6) ^c	14.8 (13.7, 15.9) ^d	< .001
Non-transit-related	21.7 (19.6, 23.9)	23.5 (19.2, 27.9)	27.8 (22.8, 32.9)	21.7 (16.9, 26.5)	.17
Nonwalking physical activity, min/d	16.0 (14.2, 17.7)	13.5 (9.9, 17.0)	11.9 (7.8, 16.0)	15.2 (11.3, 19.1)	.24

Note. CI = confidence interval. Means (95% CI) adjusted for number of assessment days; analysis includes covariates of gender, income, education, and race/ethnicity; where superscripted, not sharing a superscript denotes significant differences between groups within the same row ($P < .05$).



Note. PA = physical activity. All values are means (95% confidence intervals).

FIGURE 1—Walking and nonwalking physical activity among nontransit users (n = 419) and among transit users (n = 274) on transit days (n = 815) and nontransit days (n = 3617) in the baseline Travel Assessment and Community Study sample: King County, WA, 2008–2009.

walking, as was having a higher level of education; being obese was related to less total walking.

DISCUSSION

This study supports the growing evidence that adult transit users are more physically active than nontransit users.^{4,5} Results also suggested that higher frequency of days taking transit among those using transit at all and compared with nontransit users corresponds to the highest levels of average daily physical activity at the individual level. The most unique findings of the current study were that walking prior to and immediately after transit trips was the primary driver of observed differences in physical activity between transit users and nonusers. The differences in physical activity between transit and nontransit days among transit users were mostly explained by differences in only transit-related walking, because

no differences existed in other types of physical activity examined (other walking or nonwalking physical activity). Indeed, transit users did not appear to be more active overall or to walk more than nonusers, aside from the physical activity directly linked to their transit trips, either at the user level (transit users vs nonusers) or at the day level (transit days vs nontransit days). Physical activity also appeared to be increasing with higher frequency of transit use among transit users. When we controlled for other demographic factors, a transit day was associated with more than 12 more minutes of walking in comparison to a nontransit day.

This study found no evidence of substitution between transit-related walking and other types of physical activity either on transit days or in the form of transit users adding more physical activity on nontransit days. The average transit-related walking among transit users across all assessment days (7.6 min/d), regardless of being a transit day or not, was similar

to the difference in overall physical activity observed between transit users and nonusers across all assessment days ($46.0 - 37.6 = 8.4$ min/d). Among transit users only, the average difference between their overall physical activity on transit and nontransit days ($51.3 - 36.8 = 14.5$ min/d) was also very similar in magnitude to their transit-related walking on transit days (14.6 min/d).

There is growing interest and recognition that travel behavior is related to physical activity and is thus a health issue that affects health and chronic disease.¹² The nearly 15 minutes of physical activity per day associated directly with transit use (12.4 minutes of total walking after controlling for demographic factors) in the current study is slightly lower than the estimate by Besser and Dannenberg² (19 min/transit day) and more recently by Freeland et al.³ (18.2 min/bus transit day) but consistent with the estimated range of 12 to 15 daily minutes of reported walking associated with transit use across multiple studies.⁷ This amount of physical activity would account for half of the daily recommended level of physical activity for adults. In comparison, a recent meta-analysis by Conn et al.¹³ estimated that the average physical activity intervention (e.g., programmatic intervention targeting increases in physical activity) results in only about 2 additional minutes per day of physical activity, so the observed difference in physical activity between transit users and nonusers and between transit days and nontransit days in the current study is substantial.

Like other studies,¹⁴ the current analysis found that transit users had a lower BMI than did nontransit users. MacDonald et al.¹⁵ found a reduction in BMI with the adoption of light-rail transit commuting and a decreased likelihood of becoming obese among transit users. In the day-level analysis, current findings also indicated a difference in total daily walking between obese and nonobese adults.

Unfortunately, a low percentage of the US population uses public transportation on a regular basis, with fewer than 5% of US adults being transit users and approximately 2% of trips being made by public transportation in 2009.^{3,16} A shift toward more public transit use may be particularly influential on the physical activity of the many inactive adults not otherwise engaging in recreational physical activity.⁷

TABLE 3—Individual Predictors of Day-Level Total Walking Minutes in the Baseline Travel Assessment and Community Study Sample: King County, WA, 2008–2009

Variable	b (95% CI)
Gender (0 = female; 1 = male)	6.3* (1.8, 10.8)
Age	0.01 (-0.2, 0.2)
Race/ethnicity (0 = Hispanic or non-White; 1 = non-White Hispanic)	-3.3 (-9.7, 3.1)
Annual household income, \$ (relative to < \$50 000)	
50 000–100 000	-1.7 (-6.5, 3.1)
> 100 000	-2.2 (-7.7, 3.2)
Highest level of education (0 = < college graduate; 1 = college graduate)	6.6* (1.5, 11.6)
Employment status (relative to unemployed or retired and not working)	
Part time	2.7 (-3.4, 8.8)
Full time	2.7 (-3.2, 8.6)
Student status (0 = not a student; 1 = being a student)	3.2 (-3.5, 9.8)
Weight status (relative to BMI < 25 kg/m ²)	
Overweight (BMI = 25–30 kg/m ²)	-3.1 (-8.3, 2.1)
Obese (BMI > 30 kg/m ²)	-16.6* (-20.9, -12.3)
Used transit (0 = no; 1 = yes)	12.4* (8.7, 16.0)

Note. BMI = body mass index; CI = confidence interval.

**P* < .05.

In a small sample study in which 18 car commuters switched to public transportation for 5 days, their daily walking distance was estimated to increase by more than 1.3 kilometers, and their energy expenditure increased by 124 kilocalories per day.¹⁷ Public transportation use has other benefits, related to both higher levels of physical activity and lower carbon emissions, compared with the use of passenger cars.^{18,19} Trends are encouraging, with public transportation ridership in the United States recently growing at more than twice the population growth rate from 1995 through 2009 (34% vs 15%).²⁰ Transit-related walking also increased by 28% from 2001 to 2009 according to the National Household Travel Survey.³ Faster and more reliable transit service has been accompanied by transit ridership increases (e.g., bus rapid transit),²¹ but clearly more needs to be done to encourage transit use.

Strengths of the current study included the objective measure of physical activity and the ability to characterize time spent in physical activity as either walking or nonwalking and to identify walking that was temporally linked to transit use. However, although physical activity was objectively measured, objective and self-report measures were integrated to identify

walking episodes. Participant report of transit trips and places in travel logs also served to determine transit use. Furthermore, the sample was drawn from only 1 US metropolitan region, restricting the generalizability of findings. This study was conducted in King County, which ranks seventh in the nation's metropolitan areas for transit mode share (8.7% of trips taken by transit, less than in Philadelphia, PA, at 9.3% and more than in Baltimore, MD, at 6.2%) based on the 2009 American Community Survey.²² To focus on health effects, the current analysis was conducted at the person and day levels and not at the trip level (the standard unit of analysis in transportation statistics). Thus, the nearly 40% of our sample who reported taking at least 1 transit trip in the assessment period was difficult to compare with other transportation statistics. Census-based transit commute data provide some indication that our sample may be overrepresented by transit users in King County: almost 14% of the sample reported 4 or more transit trips in the assessment period, compared with 9.6% of the county's population who used transit to commute to work,²³ although the current analysis was not limited to commuting trips to work. Indeed, the current sample lives in census block groups where 16.7% of their

population reported using public transit for commuting, and the residential density (a proxy for transit service and ridership) of this sample's geographic area was nearly double that of the urban area of this mostly urban county (8.7 vs 4.8 housing units per acre).

With its relatively large sample, this study provided strong confirmation that transit users are more physically active than nontransit users. However, the lack of difference in activity levels observed between nontransit users and transit users on nontransit days suggested that walking to and from transit directly accounted for the higher levels of physical activity found among transit users. The ability to isolate transit walking from other types of walking reduced the likelihood of confounding by other shared variables (e.g., built environment). The ability to measure overall physical activity and, within it, to distinguish transit from non-transit-related physical activity allowed for examination of substitution effects. We found little to no evidence for substitution, and transit-related walking appeared to be added to existing physical activity among transit users. In the current study, the similarity between transit and nontransit users in demographic characteristics but significant difference in BMI and relation between weight status and walking suggested that walking to and from transit may have additional health benefits. ■

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Contributors

B. E. Saelens designed the study, conducted analyses, drafted the article, and was responsible for all aspects of the study. A. Vernez Moudon designed the study and contributed substantively to interpretation and article revision. B. Kang and P. M. Hurvitz processed and prepared the data for analysis and contributed substantively to interpretation and article revision. C. Zhou conducted analyses and contributed substantively to article revision.

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Human Participant Protection

Participants consented to participate, and the study was approved by the Seattle Children's Research Institute institutional review board.

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